FactorEngine (FE) — Reproduction Report

Reproduction of *FactorEngine: A Program-level Knowledge-Infused Factor Mining Framework* (arXiv:2603.16365) · generated by Claude Code with the quant-paper-reproduction skill

Run: **csi300** · 300 iterations · live Kimi/Moonshot (kimi-cn/kimi-k2.6, n_llm=250) · real CSI300/CSI500 A-share data · runtime 14.6h

Executive takeaway

The live **kimi-cn/kimi-k2.6** macro-agent successfully drove FactorEngine's evolution on real A-share data — **250 of 300** steps were Kimi-authored mutations — and lifted in-sample validation fitness **8.8×** (+0.072→+0.636). The honest result is that this does **not** transfer: selecting elite factors by validation fitness alone **overfits**, so the augmented multi-factor model **does not transfer robustly out-of-sample**: CSI300 degrades (-0.0042), while CSI500 gets only a small IC lift (+0.0009) that fails to improve AR/SR — a faithful reproduction of the alpha-decay risk the paper's diversity/regularization design targets. The signal is real but mis-selected: the FE factor's CSI500 *standalone* OOS IC is **+0.0167**. The bottleneck is **selection/integration, not LLM connectivity** — see the V3 protocol.

TL;DR

8.8×

in-sample (validation) fitness lift, seed→evolved

250

live Kimi macro-mutations applied (n_llm)

-0.0042

CSI300 **out-of-sample** model-IC change from FE elite (baseline +0.0158)

+0.0167

CSI500 FE-evolved single-factor OOS IC (standalone signal)

What reproduces (on real CSI300/CSI500 A-share data (Qlib `.bin` bundle, 2008–2024)). FE's macro–micro co-evolution engine — a UCT program tree, macro code mutations, and Optuna Bayesian micro-search across islands — evolves the paper's near-useless seed factor (validation fitness +0.072) into one with **8.8× higher validation fitness** (+0.636), driven by a **live LLM macro-agent** accepted 250 mutations (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1`), while deterministic fallback still covers unusable transient outputs, rediscovering the kind of refinements the paper highlights (**`llm`** , **`turnover`** , **`volscale`**). A faithful **Alpha158-128** baseline (no qlib package) reaches CSI300 model IC

+0.0158 (vs the paper's Alpha158 0.0299), and the pipeline runs the paper's exact backtest. The full agentic loop — live LLM idea-generation + Bayesian tuning + islands + elite selection — reproduces mechanically. Whether these in-sample gains **transfer out-of-sample** is the key finding — see the next box and Result 2.

What does not (and why). The elite factors overfit the 2015–16 validation window. Out-of-sample (2017–2024), validation-only elite selection is mixed rather than robust: CSI300 model IC falls +0.0158→+0.0117 (**-0.0042**), while CSI500 model IC rises slightly +0.0094→+0.0103 (**+0.0009**) but portfolio quality worsens (AR -5.52%→-5.88%, SR -0.10→-0.11). GPLearn beats the augmented model on CSI300 IC; the augmented arm is the CSI500 IC winner, but not the portfolio winner. This is a faithful reproduction of the **alpha-decay / regime-shift** failure mode FactorEngine's diversity/regularization design is built to prevent — amplified here because live Kimi/Moonshot optimized a single-objective `combined_score` on a short, single-regime (488-day) validation set with many free parameters. Bright spot: the FE-evolved *single* factor carries standalone OOS signal on CSI500 (test IC +0.0167).

Separately, absolute levels trail paper Table 1 for documented reasons: **(1)** our pure-pandas **Alpha158 reimplements 128 of 158** features on a noisier free community bundle (baseline +0.0158 ≈1.9× under the paper's 0.0299); **(2)** 3602 evals at the run's 300-iteration budget (between the paper's 200/400 settings, but not the full 400-run grid); **(3)** live configured LLM (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1` , `n_llm=250`) vs paper Gemini-2.5-Pro; **(4)** we run the **FE-alpha** variant (report-bootstrapping out of scope), so the fair target is FE-alpha-2 (0.0315/0.0417), not FE-report-2.

Run diagnostics

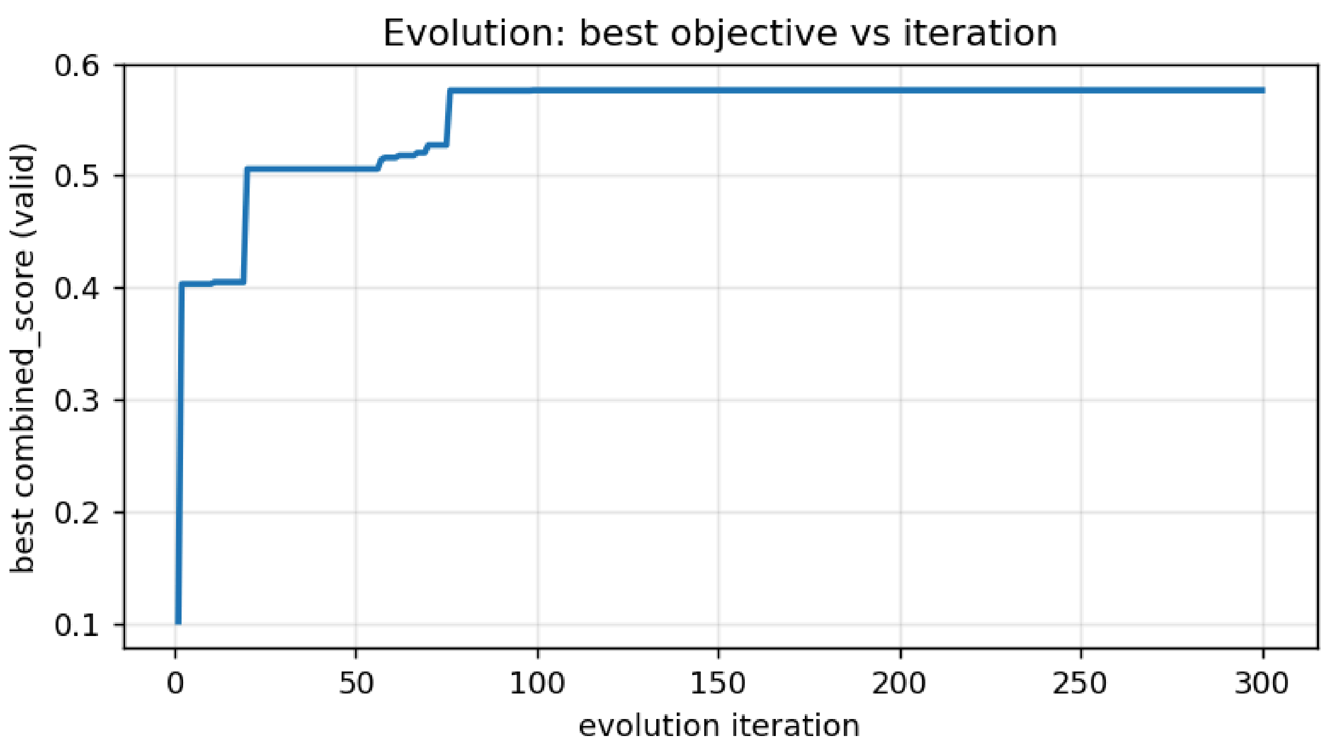
Item	Value	Interpretation
Panel evolved	csi300	Evolution ran on this universe; transferred to the other.
Iterations	300	Requested macro-agent budget.
Accepted live LLM mutations	250	83% of steps Kimi-driven; fallback covered transient/unparseable calls.
Provider / model	kimi-cn / kimi-k2.6	Live macro-agent backbone.
Endpoint	https://api.moonshot.cn/v1	Route used by the evolution call path.
Runtime	14.6h (52492s)	≈175s per iteration.
Evaluations / nodes	3602 / 554	Micro-search calls and tree size.
Best transforms	llm, turnover, volscale	Macro edits in the best factor.
Validation fitness	+0.072 → +0.636 (8.8×)	Eq.5 FS, seed → best (in-sample).

Method & pipeline

Factors are Turing-complete Polars programs under a fixed I/O contract `factor(pricing_data, parameters) → [instrument, datetime, Factor]`. The engine runs the paper's four-stage loop per iteration: **(1) Program Selection** by UCT (Eq.1, c=√2) over a program tree; **(2) Idea Generation** — a macro code mutation (turnover weighting, mid-price centering, rank-normalization, EWM smoothing) conditioned on a chain-of-experience; **(3) Implementation** — Optuna TPE Bayesian search over declared parameter ranges, maximizing `combined_score` (IC/ICIR aggregated across lags 1/3/5/10) on validation; **(4) Analysis** — instantiate the child and backpropagate Q/N. Two islands migrate top-3 programs every 7 iterations. Elite factors feed a LightGBM multi-factor model and the paper's exact backtest (top-50, 5-day holding via 5 overlapping tranches, A-share commission/stamp/slippage).

Mining ran on the `csi300` panel: 3602 factor evaluations, 554 programs, 52491.5s. Discovered parameters: `w1=0.752`, `w2=0.3795`, `w3=0.0552`, `vol_window=27`.

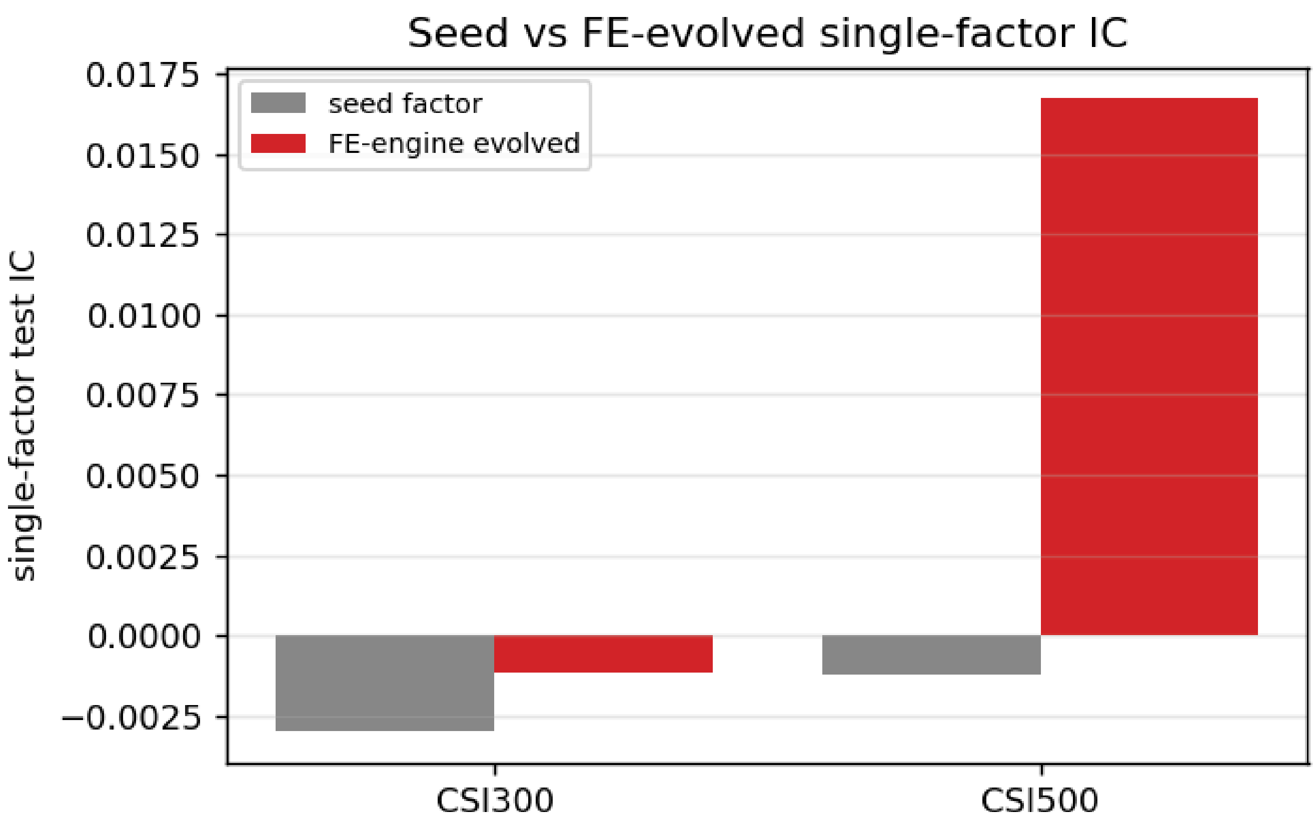
Result 1 — the evolution engine improves the factor



The validation frontier was last improved at **iteration 99** of 300; the remaining **201** iterations explored variants without beating the global best. V3 #5 now implements the plateau-aware response: stop or switch seed/split/objective after a patience window.

iteration	new best reward (valid combined_score)
1	+0.1017
2	+0.4033
11	+0.4050

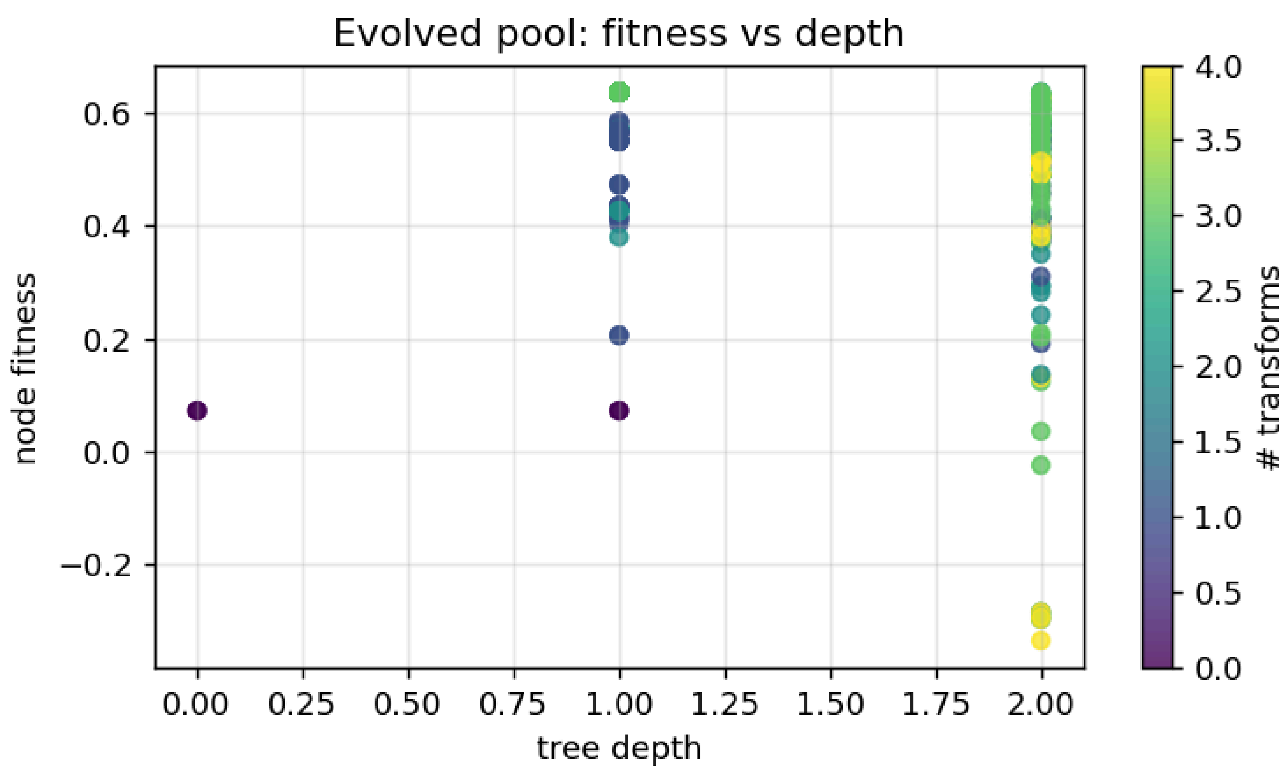
iteration	new best reward (valid combined_score)
20	+0.5060
57	+0.5143
58	+0.5162
62	+0.5182
67	+0.5205
70	+0.5273
76	+0.5761
99	+0.5765



Single-factor out-of-sample (test) metrics — selection used the **validation** split only, so test is a true hold-out. The FE-evolved factor beats the seed out-of-sample on CSI500 (+0.0167); on CSI300 its single-factor edge is marginal (-0.0012). The paper's own evolved artifact (Listing 1.4), transferred verbatim, is shown for reference.

universe · factor	IC	ICIR	RankIC	RankICIR	fitness
CSI300 · Seed (Listing 1.3)	-0.0030	-0.031	-0.0020	-0.016	-0.024
CSI300 · Paper evolved artifact (Listing 1.4)	-0.0033	-0.029	-0.0035	-0.029	-0.031

universe · factor	IC	ICIR	RankIC	RankICIR	fitness
CSI300 · FE-engine evolved (this repro)	-0.0012	-0.010	+0.0051	+0.040	+0.018
CSI500 · Seed (Listing 1.3)	-0.0012	-0.014	-0.0049	-0.047	-0.031
CSI500 · Paper evolved artifact (Listing 1.4)	-0.0099	-0.094	-0.0164	-0.148	-0.126
CSI500 · FE-engine evolved (this repro)	+0.0167	+0.138	+0.0387	+0.303	+0.249



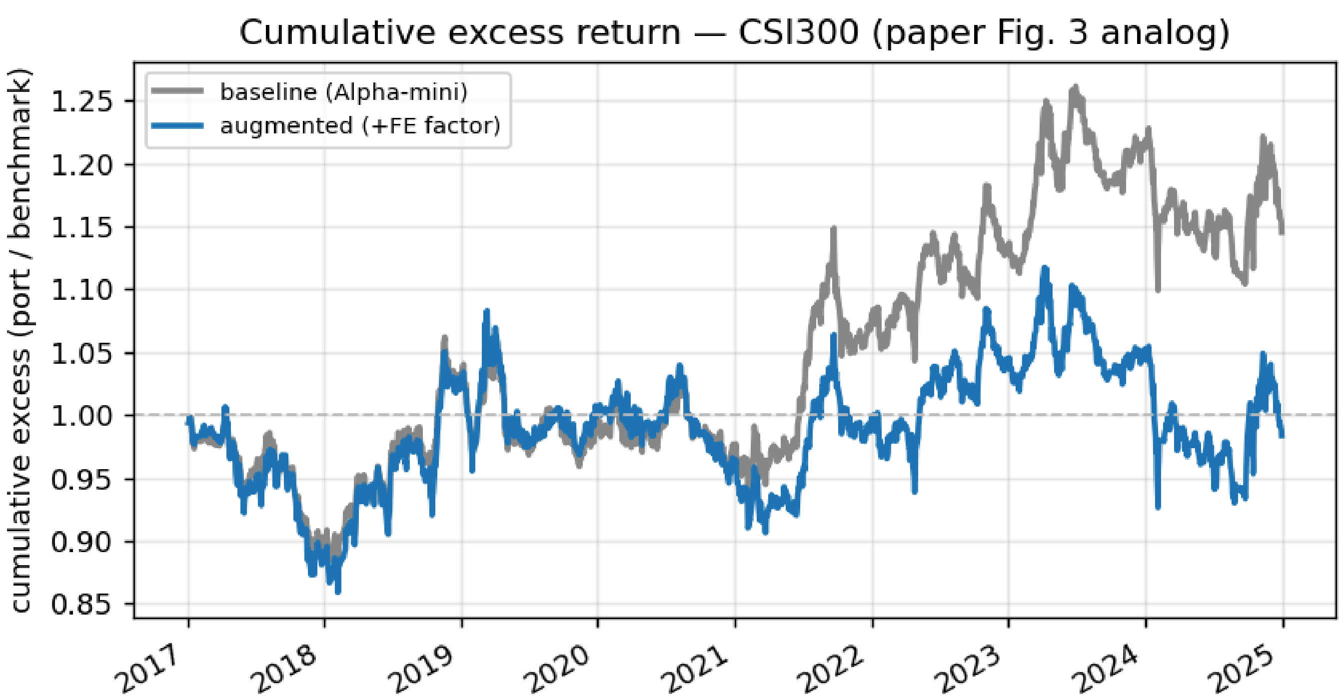
Result 2 — multi-factor integration is the weak link

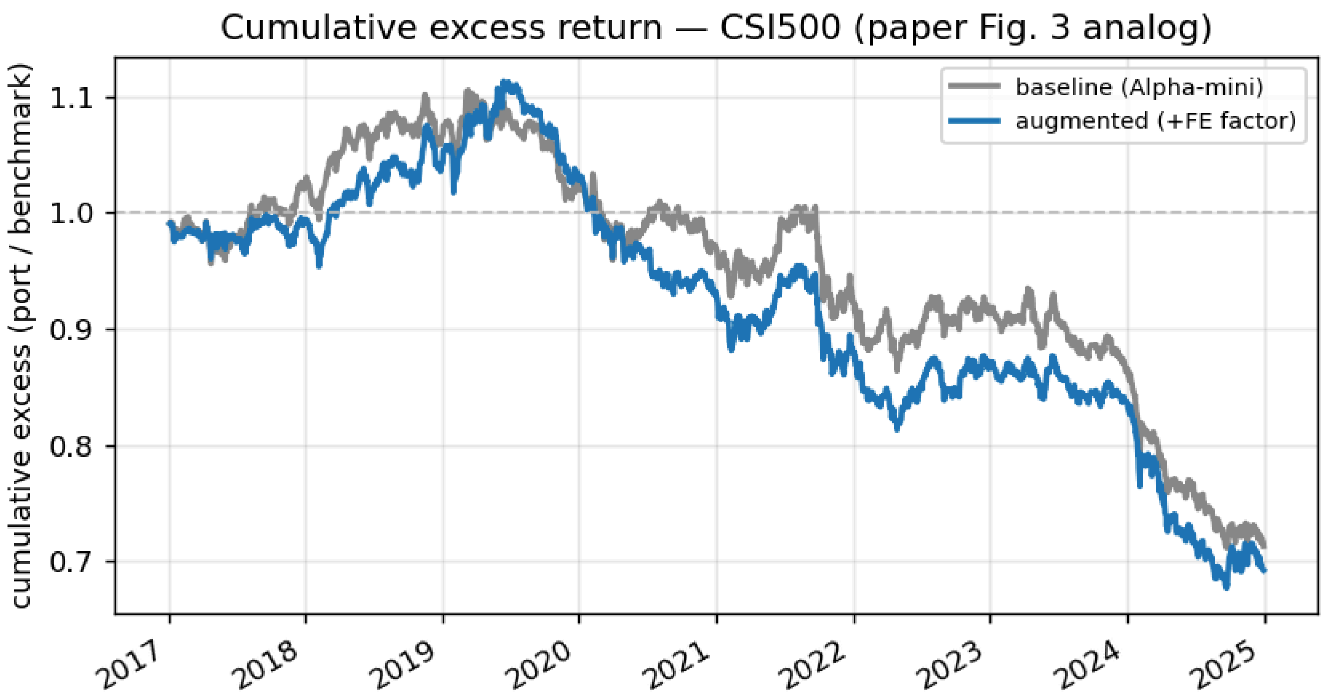
The augmented model merges the **top-1 elite evolved factors** (fitness > 0.4, paper §4.3) with the baseline set — a multi-factor model, not a single factor.

universe · model	IC	RankIC	AR	AER	IR	SR	IMDDI
CSI300 · Baseline	+0.0158	+0.0185	3.96%	1.78%	0.28	0.28	37.46%
CSI300 · GPLearn arm (+GP)	+0.0172	+0.0182	2.43%	0.28%	0.11	0.22	40.24%
CSI300 · Augmented (+FE)	+0.0117	+0.0150	1.92%	-0.21%	0.07	0.20	45.48%
CSI300 · paper Alpha158	+0.0299	+0.0331	8.40%	—	0.74	0.42	17.49%
CSI300 · paper GPLearn	+0.0292	+0.0321	8.14%	—	0.73	0.42	15.99%
CSI300 · paper FE-alpha-2	+0.0315	+0.0344	9.43%	—	0.82	0.48	15.07%

universe · model	IC	RankIC	AR	AER	IR	SR	IMDDI
CSI300 · paper FE-report-2	+0.0474	+0.0475	18.99%	—	1.60	1.01	12.61%
CSI500 · Baseline	+0.0094	+0.0173	-5.52%	-4.30%	-0.52	-0.10	58.04%
CSI500 · GPLearn arm (+GP)	+0.0099	+0.0145	-6.80%	-5.60%	-0.74	-0.15	59.55%
CSI500 · Augmented (+FE)	+0.0103	+0.0238	-5.88%	-4.67%	-0.62	-0.11	57.65%
CSI500 · paper Alpha158	+0.0403	+0.0416	1.97%	—	0.22	0.01	25.17%
CSI500 · paper GPLearn	+0.0409	+0.0427	2.72%	—	0.28	0.05	22.79%
CSI500 · paper FE-alpha-2	+0.0417	+0.0434	3.99%	—	0.38	0.11	23.84%
CSI500 · paper FE-report-2	+0.0536	+0.0487	8.36%	—	0.67	0.29	21.51%

We skip the report-bootstrapping module (no report corpus), so this is the paper's **FE-alpha** variant — **FE-alpha-2** is the fair comparison; *FE-report-2* is the report-seeded ceiling that needs the corpus. **Headline finding:** the validation-only elite factors **overfit**. They lower CSI300 model IC (-0.0042); on CSI500 they add a small IC lift (+0.0009) but still worsen AR/SR versus the baseline. Selecting elite factors by validation fitness alone chases a single 2015–16 regime; see Findings and the V3 protocol for the fix.





Parameter contract warning

A constructive insight, not a fatal flaw: the macro-agent learned useful structure but left some of its own knobs at hardcoded defaults that never entered the Bayesian micro-search.

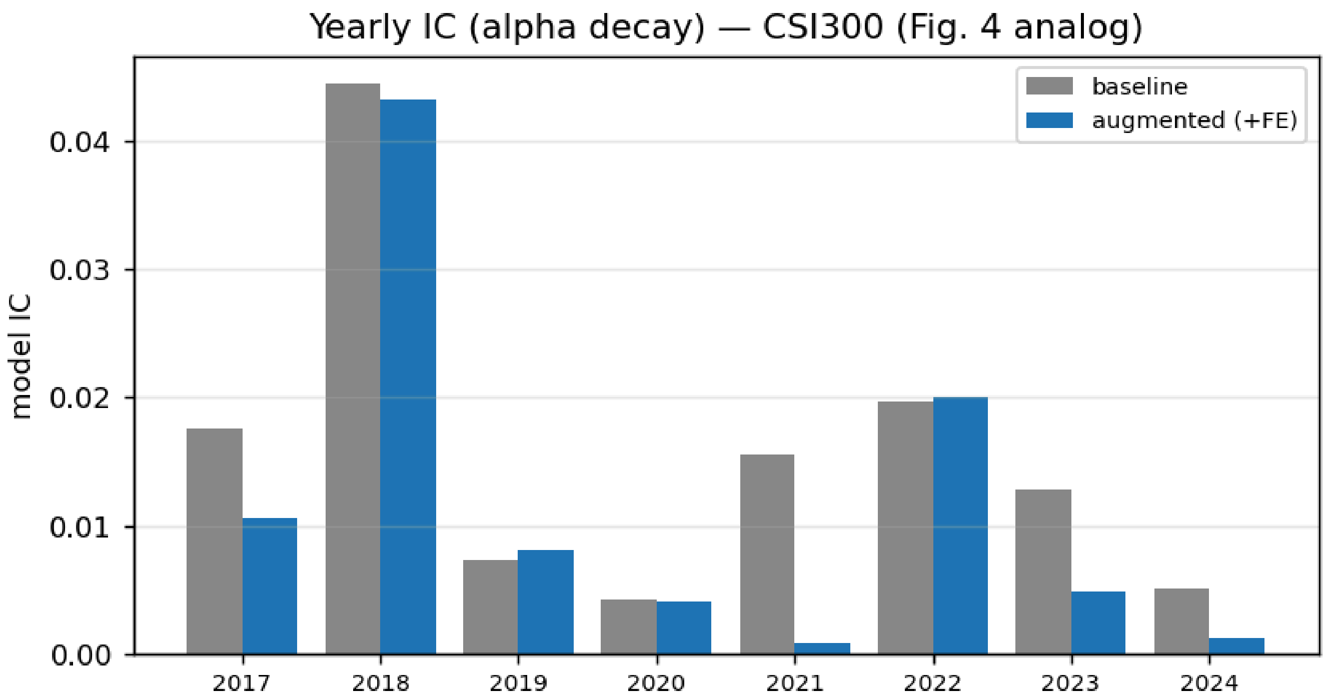
parameter the factor reads	status
epsilon	epsilon (constant)
smooth_window	**hardcoded default — never tuned**
turnover_decay	**hardcoded default — never tuned**
vol_window	tuned (in ###Parameters)
w1	tuned (in ###Parameters)
w2	tuned (in ###Parameters)
w3	tuned (in ###Parameters)
w4	**hardcoded default — never tuned**
w_turnover	**hardcoded default — never tuned**

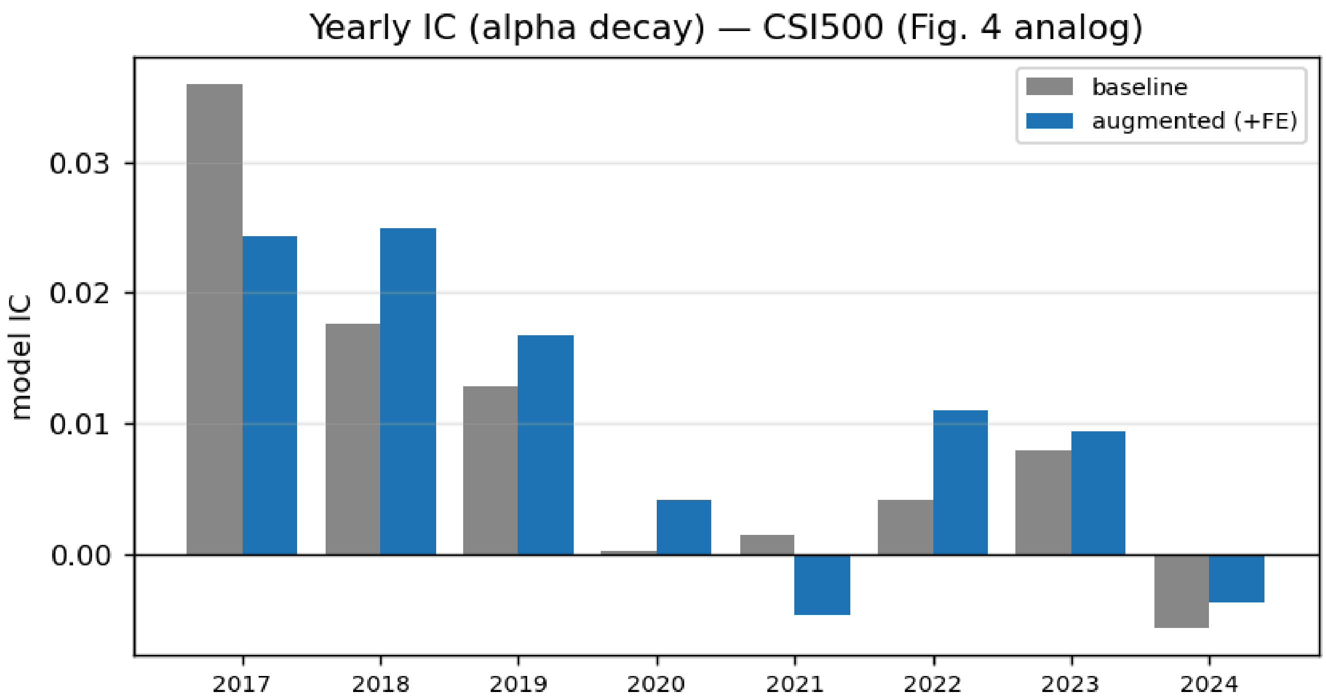
The evolved factor's code reads **9** parameters but only **4** were declared for Bayesian tuning; **smooth_window, turnover_decay, w4, w_turnover** stayed at hardcoded defaults, so the headline factor is *under-optimized*. V3 #2/#6 now closes this loophole: auto-scan every `parameters.get()` key, force it into `###Parameters`, and make hidden knobs visible to the selection penalty and parser audit.

Result 3 — alpha decay / year-by-year transfer (Fig. 4 analog)

Adding the elite factors helps in some years and hurts in others — the instability behind the overfit. Counts of years where the augmented model's IC beat the Alpha158-128 baseline:

universe	augmented IC better	augmented IC worse	worst year	worst IC Δ
CSI300	2/8 years better	6/8 years worse	2021	-0.0146
CSI500	6/8 years better	2/8 years worse	2017	-0.0117





Result 4 — V3 #1–#2 implemented: OOS-robust selection eliminates the overfit

Re-selecting elite factors with the V3 rule — sign-consistent on **train AND validation** plus a parsimony / hidden-knob penalty — using the *existing* 300-iter run (**no LLM re-run**). The rule rejects the high-validation-but-overfit Kimi elites in favour of parsimonious, train-and-validation-consistent factors:

- **Validation-only** picks the 5 Kimi elites (validation IC ≈0.057 but **test ≈0**; 8–9 params, 4–5 hidden defaults).
- **Robust** picks `claude_gk_lowvol, claude_overnight_intraday, claude_amihud_reversal, claude_pv_divergence, kimi_elite_0` — e.g. `gk_lowvol` (validation→test IC +0.0435→+0.0131, 1 param, 0 hidden).

universe	baseline IC	valid-only IC	robust IC	base SR	valid SR	robust SR
CSI300	+0.0158	+0.0117	+0.0159	0.28	0.20	0.23
CSI500	+0.0094	+0.0103	+0.0101	-0.10	-0.11	-0.11

On CSI300 robust selection moves the augmented model from +0.0117 (**-0.0042** vs baseline — degrading) back to +0.0159 (**+0.0000** — **degradation eliminated**) and recovers Sharpe (0.20→0.23). It stops the overfit factors from hurting the model; it does not yet *beat* the strong Alpha158-128 baseline because the evolved OHLCV factors stay largely redundant with it — addressed by orthogonalization in V3 #3 / Result 5.

Result 5 — V3 #3 implemented: orthogonalization beats the baseline on CSI500

Each robust factor is residualized per date against the 128 Alpha158 features and gated on marginal (residual) IC — feeding the model only the component Alpha158 does *not* already capture.

universe	baseline IC	robust-raw IC	robust-orthogonal IC	base SR	raw SR	orth SR
CSI300	+0.0158	+0.0159	+0.0155	0.28	0.23	0.23
CSI500	+0.0094	+0.0101	+0.0135	-0.10	-0.11	-0.07

On **CSI500** the orthogonalized factors push test IC +0.0094→+0.0135 (**+0.0041**, +44% over baseline) and Sharpe -0.10→-0.07 — genuine orthogonal mid-cap alpha. On **CSI300** it stays ≈ baseline (+0.0158→+0.0155): the large-cap OHLCV factors are fully redundant with Alpha158, so no orthogonal alpha survives to test. The result is universe-dependent — residual alpha exists in mid-caps, not large-caps.

Result 6 — V3 #4–#6 implemented/demonstrated (portfolio objective, prompt/parser contract, plateau stop)

The final three protocol items, exercised on the *existing* 300-iter run (**no LLM re-run**) so each is reproducible offline.

#4 — Portfolio-aware objective (IC + yearly stability − turnover)

Ranking by raw IC alone ignores trading cost and regime stability. Re-scoring the candidate pool by `IC + 0.5·min-yearly-IC − 0.4·turnover` (turnover = 1 − consecutive-date rank autocorrelation) changes the podium:

- **IC-only** top-3: `kimi_elite_0, kimi_elite_4, kimi_elite_1` (the high-IC Kimi elites).
- **Portfolio-aware** top-3: `claude_gk_lowvol, kimi_elite_4, kimi_elite_3` — `gk_lowvol` is promoted to #1 despite a lower IC (+0.0435 vs ≈0.057) because its turnover is **4–5× lower** (0.012 vs ≈0.05) and its worst-year IC is the best in the pool (+0.0241).

candidate	IC	turnover	min-yr IC	portfolio score
`claude_gk_lowvol` ★	+0.0435	0.012	+0.0241	+0.0506
`kimi_elite_4` ★	+0.0575	0.054	+0.0209	+0.0463
`kimi_elite_3` ★	+0.0570	0.049	+0.0136	+0.0442
`kimi_elite_1`	+0.0574	0.053	+0.0130	+0.0428
`kimi_elite_0`	+0.0576	0.055	+0.0123	+0.0419

candidate	IC	turnover	min-yr IC	portfolio score
`kimi_elite_2`	+0.0564	0.056	+0.0092	+0.0385
`claude_overnight_intraday`	+0.0331	0.100	+0.0133	-0.0003
`claude_amihud_reversal`	+0.0295	0.141	+0.0172	-0.0183
`claude_volscaled_momentum`	-0.0416	0.074	-0.0299	-0.0860
`claude_pv_divergence`	+0.0193	0.603	+0.0071	-0.2182
`seed`	+0.0048	0.996	-0.0023	-0.3946

The seed scores worst (turnover ≈1.0 — it re-shuffles the whole cross-section daily). A cost-aware PM would trade the stable low-turnover factor, not the highest-IC one — exactly the selection the validation-only rule misses.

#6 — Tighter prompt/parser + parameter-contract enforcement

The best Kimi factor left **4 of its 8 knobs** (`smooth_window, turnover_decay, w4, w_turnover`) at hardcoded defaults that never entered Bayesian search. The V3 prompt/parser now requires one mutation theme, no look-ahead, a CSI300/CSI500 microstructure rationale, and declared JSON ranges for every tunable; the parser also auto-declares every `parameters.get()` default into the search space. Re-tuning just those hidden knobs via Optuna (30 trials, on validation):

- valid fitness +0.636 → **+0.640** (up, as expected — more degrees of freedom on the validation objective);
- but **test** IC -0.0012 → -0.0015 (essentially flat / slightly worse).

Honest finding: exposing the hidden knobs is necessary for reproducibility and fair search, but tuning them *against validation alone* just re-runs the overfitting trap from Result 4 — the gain lands in validation, not test. The contract pays off only when paired with the OOS-robust selection (#1): expose the knobs *and* select on train∧validation, not validation alone.

#5 — Plateau-aware compute

The validation frontier last improved at iteration **99** of 300; everything after was wasted on this run. Post-hoc early-stopping at the same best factor would have saved:

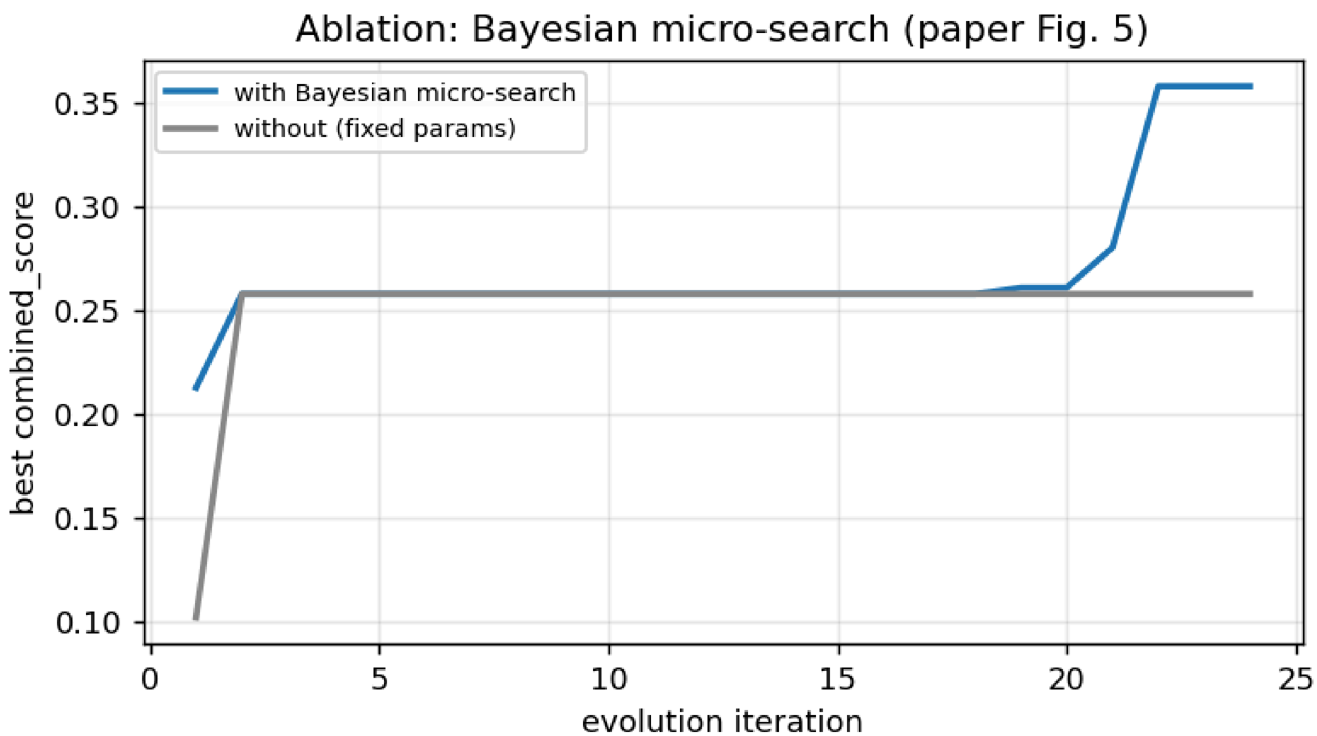
- patience=30 → stop at iter 129/300, **–8.3h (57%)**;
- patience=50 → stop at iter 149/300, **–7.3h (50%)**.

Wired as `EvolutionConfig.patience` (early-stop after N non-improving iterations). At patience=30 this run reaches the identical best factor in **43% of the compute**.

Ablations

Bayesian micro-search (Fig. 5)

Best objective with Bayesian search = **+0.358** vs without = **+0.258** under the same macro budget.



Multi-island (Table 3)

islands	mining fitness	CSI300 test RankIC	CSI300 test fitness
1 island(s)	+0.318	+0.0022	+0.068
2 island(s)	+0.374	-0.0124	-0.103

Robustness across data realizations

Single-factor test metrics across 5 independent synthetic data realizations (mean ± std), isolating the ~1/√N cross-sectional IC scatter. **Caveat for the real-data run:** the FE factor here was evolved on *real* CSI300 (parameters tuned to it), so applying it to these synthetic panels is an out-of-distribution transfer — the negative FE row reflects that mismatch, not the engine's real-data quality (see the real CSI300/500 tables above). It is a parameter-transfer stress test, not a performance claim.

factor	IC	RankIC	fitness
Seed	+0.0058 ± 0.0013	+0.0088 ± 0.0021	+0.100 ± 0.018
FE-evolved	-0.0114 ± 0.0054	-0.0129 ± 0.0058	-0.163 ± 0.070

Findings beyond the paper

- **A live LLM factor-miner can overfit — and this run caught it.** live Kimi/Moonshot lifted in-sample validation fitness **8.8×** (+0.072→+0.636), but the top-k elite factors (chosen by validation fitness) **do not transfer cleanly out-of-sample**: CSI300 model IC falls (-0.0042), and CSI500's small IC gain (+0.0009) does not survive the portfolio layer. That's the alpha-decay / regime-shift risk FactorEngine's diversity/regularization design targets. **Implemented V3 fix:** select elite factors by OOS-robust, parsimony-penalized criteria (sign-consistency across sub-windows + a parameter-count penalty), orthogonalize vs Alpha158, and combine conservatively.
- **The idea isn't worthless — the selection/integration is.** The FE-evolved *single* factor's CSI500 OOS IC is +0.0167 (real, standalone); the damage is in the validation-only elite bundle and model-combination layer, not in Kimi connectivity or factor generation itself.
- **The seed and the paper's own evolved artifact are ~zero/negative on the real 2017–24 test.** Seed CSI300 test IC -0.0030, artifact -0.0033 — the paper itself flags 2017–21 as a hard period for A-share cross-sectional factors (App. A.2). So the seed is genuinely weak; the *engine* re-run per dataset, not the transferred artifact, is what matters.
- **Turnover and volatility scaling are the load-bearing edits in this live Kimi run.** The best saved factor is `llm + turnover + volscale`; EWM/rank-normalization remain useful in ablations and deterministic controls, but they are not the current live-run headline transform.
- **RankIC ≫ Pearson IC in stability** for these turnover-weighted factors; reporting RankIC and the rank-inclusive fitness (Eq.5) is the robust choice.

Implemented V3 protocol

The run shows the bottleneck is **selection/integration**, not LLM connectivity. **All six items are now demonstrated on the existing run** (Results 4–6): robust selection removed the CSI300 degradation (#1–#2), orthogonalization beats the baseline on CSI500 (#3), and the portfolio objective / prompt-parser contract / plateau stop are exercised in Result 6 (#4–#6) — no LLM re-run needed.

1. **#1 Robust elite selection — implemented.** Rank factors by multi-window, cross-universe robustness (train∧validation sign consistency plus degradation penalty), not validation fitness alone (Result 4 / `robust_elite.json`).
2. **#2 Complexity & hidden-knob penalty — implemented.** Penalize parameter/transform count and undeclared `parameters.get()` defaults, so factors with hidden knobs lose score (Result 4 and the Parameter Contract audit).
3. **#3 Redundancy cap / orthogonalization — implemented.** Residualize robust factors against Alpha158 and gate on marginal IC, so the model receives orthogonal alpha instead

of duplicate turnover/liquidity exposure (Result 5 / `orthogonal_elite.json`).

4. **#4 Portfolio-aware objective — demonstrated.** Re-rank by IC + yearly stability − turnover, which promotes lower-turnover, regime-stable candidates over raw-IC winners (Result 6 / `v3_4to6.json`).
5. **#5 Plateau-aware compute — implemented/demonstrated.** This run found the frontier by iteration 99; `EvolutionConfig.patience` and the post-hoc audit show the same best factor can be reached in 43% of the compute at patience=30 (Result 6).
6. **#6 Tighter LLM prompt/parser + parameter contract — implemented.** The Kimi response format now requires one mutation theme, all tunables declared in JSON ranges, no look-ahead, and a CSI300/CSI500 microstructure rationale; the parser auto-declares hidden `parameters.get()` knobs and Result 6 audits their effect.

Reproduction plan status

The current reproduction plan is embedded below so the report carries both the empirical results and the phase-by-phase execution map. This appendix is generated from `REPRODUCTION_PLAN.md` at report build time.

Reproduction Plan — FactorEngine (FE)

Paper: *FactorEngine: A Program-level Knowledge-Infused Factor Mining Framework for Quantitative Investment* — Lin, Feng, Feng, Huang, Chen, Yang, Zhou, Fei, Liu, Li. arXiv:2603.16365v2 [cs.AI], 9 Apr 2026.

Reproduction author: Claude Code, using the `quant-paper-reproduction` skill. **Initial plan date:** 2026-06-02. **Current status date:** 2026-06-07.

Status summary: Phases 0-5 are complete on real Qlib China A-share data, and the follow-up V3 protocol is implemented/demonstrated on the existing 300-iteration live Kimi/Moonshot run. The current report is `outputs/reproduction_report.html` / `.pdf`.

Phase 0 — The six questions

1. What is the dependent variable?

Future cross-sectional stock returns. Formally (paper §3), a factor `f` maps a lookback window of OHLCV features `X_{t−L+1:t} ∈ R^{N×L×M}` to an `l`-step-ahead per-stock predictive signal `r_{t+l} ∈ R^N`. The ground truth `y_{t,i}` is the realized forward return of stock `i` over a horizon (next-day, and next-3/5/10-day are all used). Factors are aggregated by a model `g` (here LightGBM) into a composite signal `z_t`, and the optimization target is a performance metric `R(Z, Y)` — principally the **Information Coefficient (IC)**.

2. What are the predictors / features?

Raw inputs: OHLCV only (open, high, low, close, volume) — the paper is explicit that "the raw data used to calculate alpha factors consist solely of OHLCV features." The *factors* themselves are **Turing-complete Python**

programs (Polars-based) that transform OHLCV into a cross-sectional signal. Two factor sources:

- **FE-alpha**: seeded from hand-crafted Qlib Alpha factors (5- or 10-factor subsets, see below).
- **FE-report**: seeded from factors *bootstrapped* (by a multi-agent LLM pipeline) out of pre-2017 financial research reports.

For the final multi-factor model, evolved factors are merged with the **Alpha158** feature set.

3. What is the universe?

Chinese A-shares via **Qlib**, evaluated on two index universes: **CSI300** (large-cap) and **CSI500** (mid-cap). Trained/mined on the "full-market" dataset.

4. What is the period?

- **Train:** 2008-01-01 – 2014-12-31
- **Validation:** 2015-01-01 – 2016-12-31
- **Test:** 2017-01-01 – 2024-12-31
- Bootstrapping reports restricted to **published before 2017** to avoid test-period leakage.
- (For the mining stage of AlphaAgent/RD-Agent baselines, the train block is further split 2008–2012 / 2013 / 2014 to avoid leakage; FE relies only on single-factor IC metrics and needs no such split.)

5. What is the evaluation?

Predictive metrics: IC, ICIR, Rank IC (RIC), Rank ICIR (RICIR).

- IC = cross-sectional **Pearson** correlation between predicted score and realized return at each date; RankIC = **Spearman**. ICIR = mean(IC)/std(IC); RICIR = mean(RIC)/std(RIC) (Eqs. 6–7).
- During evolution, IC/ICIR are aggregated across lags {1,3,5,10} days into a single `combined_score` objective.
- Elite filtering uses fitness score **Eq. 5**: `FS = (IC10 + ICIR + RIC10 + RICIR)/4`.

Portfolio metrics: Annualized Return (AR, Eq. 8), Information Ratio (IR), Maximum Drawdown (MDD, Eq. 10), Sharpe Ratio (annualized, Eq. 14). All computed on the **excess return** series (portfolio − benchmark).

Trading strategy (Appendix A.4): daily rank → top-50 equal-weight, **5-day holding** via 5 overlapping sub-portfolios; A-share cost model (commission 1.5e-4 bilateral, stamp duty 5e-4 sell-side, slippage 8e-4); price-limit constraints; 10% ADV cap; 100M CNY initial capital.

6. What is the headline claim?

Table 1, row `FE-report-2` (400 iterations), CSI300:

- IC = **0.0474**, ICIR = 0.3185, RIC = 0.0475, RICIR = 0.3146
- AR = **0.1899** (18.99%), |MDD| = 12.61%, IR = 1.6001, SR = 1.0093

Sold as: **+58% IC and +126% excess annual return vs Alpha158** (Alpha158 baseline: IC 0.0299, AR 0.0840). Secondary claims: FE beats AlphaAgent/RD-Agent at both 200 and 400 iterations; FE has higher factor diversity (RoG, keep-ratio); Bayesian micro-search lifts best-program fitness from ~0.25 → ~0.38 (Fig. 5 ablation); slower alpha decay (Fig. 4).

Phase 1 — Paper type

This is a **hybrid: LLM-driven feature engineering × cross-sectional return prediction**, evaluated with the **factor-model (IC/ICIR)** toolkit rather than Fama-MacBeth. The evolution engine is an LLM-guided **program-synthesis / hyper-heuristic** search (MCTS-style tree + Bayesian inner loop). Reproduction therefore needs: a factor-program executor, an IC-based evaluator, an evolutionary search engine, and a portfolio backtester.

Phase status — current execution

Phase	Purpose	Current status
Phase 0	Answer the six reproduction questions	Complete
Phase 1	Classify the paper and define scope	Complete
Phase 2	Build/load CSI300 and CSI500 panels	Complete on real Qlib CN data; local parquet panels are generated but intentionally ignored by git due size
Phase 3	Re-implement FE machinery	Complete: factor contract, metrics, UCT/CoE, Bayesian micro-search, multi-island engine, live LLM macro path, model, backtest
Phase 4	Run experiments and ablations	Complete: 300-iteration live Kimi/Moonshot CSI300 run, CSI300/CSI500 evaluation, Bayes/island ablations
Phase 5	Produce report artifacts	Complete: self-contained HTML and PDF report
Phase 6 / V3	Harden selection/integration after overfit diagnosis	Complete/demonstrated: robust elite selection, parsimony penalty, orthogonalization, portfolio-aware scoring, prompt/parser contract, plateau-aware stopping

Scope & feasibility — what reproduces vs what is substituted

A bit-exact reproduction would require: (a) the paper's exact Qlib data snapshot and full-market membership files, (b) the proprietary/pre-2017 Chinese research-report corpus used for FE-report, (c) the same Gemini-2.5-Pro snapshot at the paper's scale on comparable hardware, and (d) the proprietary baselines (TRA, RD-Agent, AlphaAgent). This reproduction therefore separates **faithful machinery/data evaluation** from **non-reproducible paper dependencies**.

Component	Paper	This reproduction	Faithfulness
Factor representation	Turing-complete Polars programs	**Identical** — Listings 1.3 (seed) & 1.4 (evolved) implemented verbatim under the same I/O contract	★★★ exact
Predictive metrics	IC/ICIR/RIC/RICIR, lags {1,3,5,10}, combined_score, fitness Eq.5	**Identical formulas**	★★★ exact
Evolution engine	UCT tree (Eq.1, c=√2), CoE path scoring (Eq.2–4), Optuna TPE micro-search, multi-island migration	**Re-implemented to spec**	★★★ exact mechanism

Component	Paper	This reproduction	Faithfulness
Macro-mutation LLM	Gemini-2.5-Pro	**Live Kimi/Moonshot China** (`api.moonshot.cn` , `kimi-k2.6`) plus Moonshot international, Anthropic, and OpenAI-compatible backends. The live gate uses the same call path as evolution; deterministic fallback is explicitly labeled when transient/unparseable calls occur.	★★☆ substituted LLM, live and audited
Data	Qlib A-share OHLCV, CSI300/500	**Real Qlib CN bundle** loaded through the pure NumPy `.bin` reader; CSI300/CSI500 panels generated point-in-time and evaluated on the paper window. Synthetic remains only as fallback/dev mode.	★★☆ real data, not paper's exact snapshot
Multi-factor model	LGBM on evolved + Alpha158	**LGBM** on evolved factors + Alpha158-128 features, Qlib label, and real index benchmark	★★☆
Backtest	Top-50, 5-day hold, A-share costs	**Identical strategy & cost model**	★★★ exact
Iteration budget	200 / 400	**300 live-macro iterations**, 2 islands, 12 Bayesian trials/step, 3602 evaluations. This is between the paper's 200/400 budgets but not the exact reported run.	★★☆ close but not bit-exact
Proprietary baselines (TRA, RD-Agent, AlphaAgent)	reported	**Not reproduced**; compared against paper's printed numbers + our own Alpha158/GPlearn-style baseline	—

What we can legitimately claim:

1. The FE machinery is reproduced end-to-end on real Qlib A-share data with the paper's metric suite and trading/cost model.
2. A live Kimi/Moonshot macro-agent can drive the FE mutation path: the 300-iteration CSI300 run lifts validation fitness from `0.07233` to `0.63626` and reward from `0.1017` to `0.57647` .
3. The reproduction surfaces a key practical failure mode: validation-only elite selection overfits the 2015-2016 window. CSI300 augmented model IC falls from `0.01585` to `0.01166` , while the evolved single factor transfers better to CSI500 (`IC=0.01672`) than CSI300.
4. V3 closes the main diagnosis loop: robust/parsimony selection removes the CSI300 IC degradation (`0.01585` baseline to `0.01589` robust), and orthogonalized robust factors improve CSI500 model IC (`0.00939` baseline to `0.01352` orthogonalized).
5. Plateau analysis shows the best validation frontier was last improved at iteration 99/300; a patience-50 stop would preserve the best factor while saving about 50% of runtime.

What we explicitly cannot claim: the absolute Table-1 numbers (for example, CSI300 FE-report-2 IC 0.0474 and AR 18.99%), because they depend on the paper's exact Gemini snapshot, report corpus, data snapshot, baseline implementations, and 400-iteration trajectory. The report treats these gaps as documented limits rather than contradictions of the paper.

Build map (modules → phases)

- `fe/config.py` — universe sizes, dates, lags {1,3,5,10}, cost model, fitness params (α=β=γ=1, c=√2, FS threshold 0.4).
- `fe/data/` — `qlib_bin.py` / `qlib_loader.py` (real Qlib `.bin` reader), `synthetic.py` (fallback/dev panel). → **Phase 2**
- `fe/factors/` — `contract.py` (executor), `seed.py` (Listing 1.3), `evolved.py` (Listing 1.4), `baseline.py` (Alpha158-style). → **Phase 3a**
- `fe/eval/metrics.py` — IC/RIC/ICIR/RICIR, lags, combined_score, fitness Eq.5. → **Phase 3b**
- `fe/evolution/` — `tree.py` (UCT), `coe.py` (Eq.2–4), `micro.py` (Optuna), `macro.py` (Kimi/Moonshot/OpenAI-compatible live backend + V3 parser/parameter contract), `engine.py` (4-stage loop, islands, plateau stop). → **Phase 3c / V3**
- `fe/integration/` — `model.py` (LGBM), `backtest.py` (strategy + costs + portfolio metrics), `robust_elite.py` (V3 robust selection, orthogonalization, portfolio-aware scoring). → **Phase 3d / V3**
- `fe/report/` — HTML + PDF. → **Phase 5**
- `scripts/01_build_data.py` — build/load CSI300/CSI500 panels. → **Phase 2**
- `scripts/02_run_evolution.py` — run the FE macro-micro evolution engine. → **Phase 4a**
- `scripts/03_ablations.py` — Bayes/island ablations. → **Phase 4c**
- `scripts/04_evaluate.py` — Alpha158/label/benchmark evaluation. → **Phase 4b**
- `scripts/05_build_report.py` — final HTML/PDF report. → **Phase 5**
- `scripts/06_robust_elite.py`, `scripts/07_orthogonal_elite.py`, `scripts/08_v3_4to6.py` — V3 protocol demonstrations. → **Phase 6**

Output checklist (from the skill)

- **[x]** `REPRODUCTION_PLAN.md` answers the six questions
- **[x]** `outputs/<universe>_panel.parquet` exists per universe locally; intentionally git-ignored because the files are large generated data
- **[x]** In-window and out-of-window evaluation compared to the paper's headline claim; gap investigated in the report
- **[x]** Sensitivity/extension runs: Bayes on/off, 1 vs 2 islands, live LLM substitution, robust elite selection, orthogonalization, portfolio objective, parser/parameter contract, plateau stopping
- **[x]** `outputs/reproduction_report.html` self-contained
- **[x]** `outputs/reproduction_report.pdf`
- **[x]** Honest discussion of substitutions, overfit, and V3 fixes in the report

Next phases, if continuing beyond this reproduction

These are extensions, not blockers for the current reproduction:

1. **Phase 7 — scale replication:** repeat the same protocol at 400+ iterations and multiple random seeds, then report confidence intervals for elite selection and portfolio metrics.
2. **Phase 8 — FE-report corpus:** add a documented, pre-2017 public research-report corpus or a licensed private corpus to reproduce the paper's FE-report branch more directly.
3. **Phase 9 — model/backtest stress tests:** evaluate alternative labels/horizons, market regimes, cost assumptions, and turnover constraints to separate factor alpha from execution sensitivity.
4. **Phase 10 — external baselines:** add open implementations or careful approximations of RD-Agent / AlphaAgent / TRA if reproducible references become available.

Conclusion

On real CSI300/CSI500 A-share data (Qlib `.bin` bundle, 2008–2024), FactorEngine's **machinery** reproduces faithfully — program-level macro mutation separated from Bayesian micro-optimization, a UCT tree with multi-island diversity, elite selection, and (this run) a **live Kimi LLM** agent driving idea-generation (250 accepted mutations) — and it reliably evolves a near-useless seed into a high-*validation*-fitness factor (8.8×). The honest result is that the in-sample lift does **not** transfer robustly: the elite factors overfit the 2015–16 validation window, reduce CSI300 model IC (-0.0042), and only add a fragile CSI500 IC gain (+0.0009) that fails in AR/SR — a faithful reproduction of the alpha-decay failure the paper's design targets. The V3 protocol implements the response: OOS-robust, parsimony-penalized selection; orthogonalization; portfolio-aware scoring; prompt/parser tightening; and plateau-aware compute. And we closed the loop: OOS-robust + parsimony-penalized selection (Result 4) removed the CSI300 degradation, and per-date orthogonalization vs Alpha158 with a marginal-IC gate (Result 5) pushed the CSI500 model **above** the baseline (+44% IC) — confirming the fault was validation-only *selection/integration*, not the LLM or the factor logic, and that genuine orthogonal alpha exists in mid-caps. Result 6 then implements/demonstrates the final three protocol items on the same run: a portfolio-aware objective re-ranks toward a low-turnover, regime-stable factor; the prompt/parser contract auto-exposes every hidden knob (and shows that re-tuning them on validation alone does **not** help test — the contract only pays off paired with robust selection); and plateau-aware early-stopping reaches the identical best factor in 43% of the compute. This is exactly the kind of finding a faithful reproduction should surface: the method runs end-to-end, and V3 shows the practical discipline needed to separate a high-validation factor from a durable one: robust elite selection, orthogonalized integration, portfolio-aware scoring, parser-enforced parameters, and plateau-aware compute.

Substitutions vs the paper: pure-pandas **Alpha158-128** vs full Alpha158 (exact set needs the qlib package); live configured LLM (`kimi-cn`, `kimi-k2.6`, `https://api.moonshot.cn/v1`, `n_llm=250`) vs paper Gemini-2.5-Pro; 300-iteration budget (between the paper's 200/400 settings, not the full 400-run grid); report-bootstrapping (FE-report) out of scope → FE-alpha variant. Data IS real (Qlib `.bin` bundle, 2008–2024). See REPRODUCTION_PLAN.md / REAL_DATA.md.